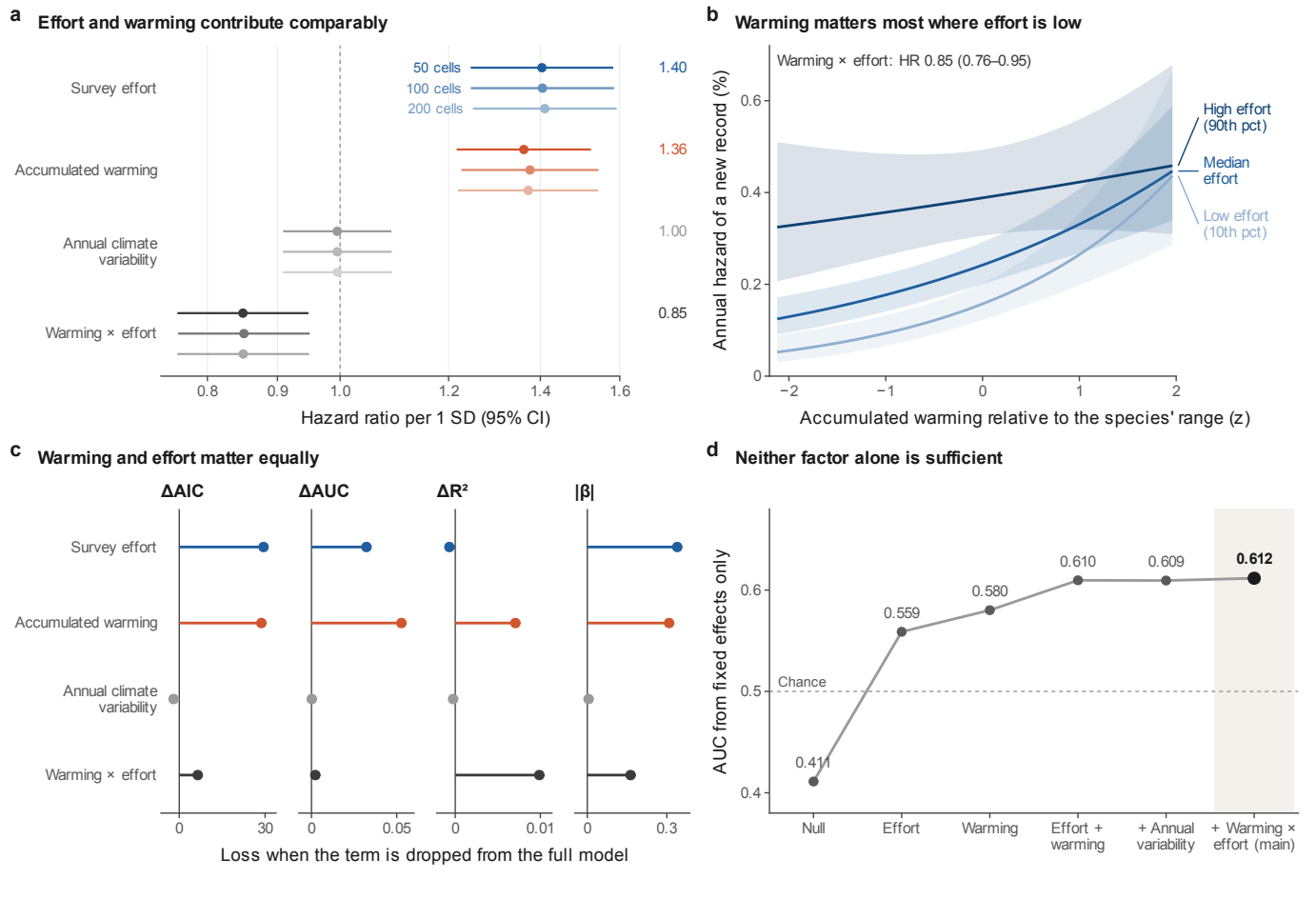

a
Effort and warming contribute comparably
50 cells
100 cells
200 cells
1.40
1.36
1.00
0.85
Survey effort
Accumulated warming
Annual climate variability
Warming × effort
0.8
0.9
1.0
1.2
1.4
1.6
Hazard ratio per 1 SD (95% CI)
b
Warming matters most where effort is low
Warming × effort: HR 0.85 (0.76–0.95)
High effort (90th pct)
Median effort
Low effort (10th pct)
Annual hazard of a new record (%)
0
0.2
0.4
0.6
−2
−1
0
1
2
Accumulated warming relative to the species' range (z)
c
Warming and effort matter equally
ΔAIC
ΔAUC
ΔR²
|β|
Survey effort
Accumulated warming
Annual climate variability
Warming × effort
0
30
0
0.05
0
0.01
0
0.3
Loss when the term is dropped from the full model
d
Neither factor alone is sufficient
AUC from fixed effects only
0.4
0.5
0.6
Chance
0.411
0.559
0.580
0.610
0.609
0.612
Null
Effort
Warming
Effort + warming
+ Annual variability
+ Warming × effort (main)